Standard sum of products (SOP)

Standard product of sums (POS)

minterm: combination in truth table

maxterm: combination in truth table

half-adder: two one-bit binary inputs, a carry out and a sum

full-adder: two one-bit binary inputs a carry in, a carry out and a sum.

half-adder



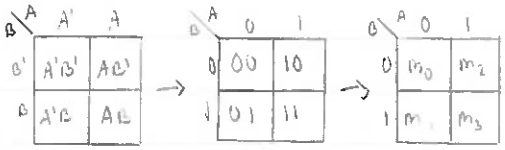
input		output	
A	B	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Chapter 3 - Level minimization

K-map (Karnaugh map):

- four or less variables
- each square is a minterm
- 2^n squares

* don't cares can be 1's or 0's



Prime implicant: group of maximum adjacent squares.

Essential prime implicant: group that has at least one square that isn't covered by another group.

Optional prime implicant: group that has all its squares covered by another group.

multi-level implementation (NAND)

1. convert all AND gates to NAND
2. convert all OR gates to NAND
3. Check all the bubbles in the diagram.

Two-level implementation (NAND)

1. Simplify the function and express it in SOP
2. Draw a NAND gate for each product term that has at least 2 literals.
3. Draw a single gate using the AND invert or the invert OR in the second level, with inputs coming from outputs of first level gates.
4. A term with 1 literal requires an inverter in the first level. * If the single literal is complemented, it can be connected directly to an input of the second level NAND gate.

NOR gate implementation

- function needs to be simplified into POS
- changing OR gates to NOR gates
- changing AND gates to NOR gates
- single literal going into the second-level gate must be complemented